

System Monitoring and Management

Objectives

After completing this lesson, you should be able to:

- Use the `sosreport` utility
- Use the `iostat`, `mpstat`, `vmstat`, `sar`, `top`, `iostat`, and `strace` utilities
- Use the `netstat` and `tcpdump` utilities
- Use the Wireshark network analyzer GUI
- Use the OSWatcher (OSWbb) tool
- Use OSWatcher Analyzer (OSWbba)
- Describe Oracle Enterprise Manager Cloud Control
- Describe Spacewalk



sosreport Utility

- The `sosreport` utility:
 - Collects debugging information about a system
 - Stores the information in a compressed file in `/var/tmp`
- Run the tool as follows:

```
# sosreport
...
Please enter your first initial and last name...
Please enter the case number...:
```

- `sosreport` uses plug-ins. Options exist to manage plug-ins:
 - `-l`: List the status of all available plug-ins.
 - `-n PLUGNAME`: Do not load specified plug-in(s).
 - `-e PLUGNAME`: Enable specified plug-in(s).

iostat Utility

- The `iostat` utility:
 - Reports CPU and I/O statistics
 - Is used during performance analysis to balance I/O load
- The `iostat` utility report has the following sections:
 - CPU utilization
 - Device utilization
- Execute `iostat` continuously at a specific *interval*, up to *count* times:

```
# iostat interval count
```

- For example, to run `iostat` every 10 seconds for 5 times:

```
# iostat 10 5
```

Example output of iostat

```
# iostat
```

```
•  avg-cpu:  %user   %nice %system %iowait  %steal   %idle
•           0.08    0.00    0.04    0.56    0.61       98.70

•  Device:            tps    kB_read/s    kB_wrtn/s    kB_read
   kB_wrtn
•  xvda                0.96          6.88          4.91      615926
   439527
•  xvdb                0.01          0.18          0.02      15912
   2052
•  xvdd                0.01          0.18          0.02      15928
   2052
•  scd0                0.01          0.06          0.00       5080
   0
```

mpstat Utility

- The `mpstat` utility:
 - Collects and displays performance statistics for all CPUs
 - Is used during performance analysis to determine CPU utilization
- Use the `-P ALL` option to include average usage of all CPUs.
- Execute `mpstat` continuously at a specific *interval*, up to *count* times.

```
# mpstat -P ALL
```

- For example, to run `mpstat` every 2 seconds for 5 times:

```
# mpstat interval count
```

```
# mpstat 2 5
```

Example output of mpstat

```
# mpstat
```

```
07:25:58 AM  CPU      %usr   %nice    %sys %iowait    %irq   %soft
%steal  %guest  %gnice   %idle
07:25:58 AM  all        0.06    0.00        0.03    0.52        0.00
0.00    0.60        0.00    0.00    98.79
```

vmstat Utility

- The `vmstat` utility:
 - Monitors system memory usage
 - Is useful for detecting shortages of physical memory
- The `vmstat` report has six sections:
 - `Processes`: Number of processes in wait or sleep states
 - `Memory`: Amount of memory free and amount used for virtual memory, buffers, and cache
 - `Swap`: Number of page-ins and page-outs
 - `IO`: Number of blocks received and sent
 - `System`: Number of interrupts and context switches
 - `CPU time`: Percentages for user, kernel, idle, iowait, and stolen
- **Recommended:** Run the utility with a delay interval:

```
# vmstat 5
```


sar Utility

- Provided by the `sysstat` package:
 - `sar`: Collects and displays ALL system activities statistics
 - `sadc`: Is the `sar` back-end tool that does the data collection
 - `sa1`: Is a script that runs `sadc` and stores system activities in a binary data file. `sa1` runs from `cron`.
 - `sa2`: Creates daily summary of the collected statistics. `sa2` runs from `cron`.
 - `pidstat`: Reports statistics based on the process ID (PID)
 - `cifsiostat`: Generates CIFS statistics
- Many options exist for `sar`:
 - `-A`, `-r`, `-b`, `-B`, `-d`, `-S`, and more
- You can specify *interval* and *count* parameters.

top Utility

- The `top` utility monitors system processes in real time.
- The upper section of the `top` output displays load averages, number of running and sleeping tasks, and overall CPU and memory usage.
- The lower section has a sorted list of processes, owner, running time, and CPU and memory usage.
- `top` sorts the list by most CPU-intensive tasks and refreshes the list every three seconds by default.
- `top` provides an interactive interface for manipulating processes:
 - `h` or `?`: Display the help screen.
 - `F` or `f`: Display the field management screen.
 - `i`: Toggle the display of all tasks or just active tasks.
 - `q`: Quit.

Example output of top

top

top - 08:30:45 up 1 day, 23:12, 2 users, load average: 0.00, 0.01, 0.05

Tasks: 127 total, 2 running, 125 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0/0.3 0[]

KiB Mem : 2045360 total, 871460 free, 247752 used, 926148 buff/cache

KiB Swap: 4281340 total, 4281340 free, 0 used. 1729276 avail Mem

	PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
•	754	root	20	0	0	0	0	S	0.3	0.0	0:00.58	[kworker/0:2]
•	1	root	20	0	128172	8288	5452	S	0.0	0.4	0:10.10	/usr/lib/systemd/systemd --swit+
•	2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	[kthreadd]

iotop Utility

```
Total DISK READ :      0.00 B/s | Total DISK WRITE :      0.00 B/s
Actual DISK READ:      0.00 B/s | Actual DISK WRITE:      0.00 B/s
  TID  PRIO  USER      DISK READ  DISK WRITE  SWAPIN      IO>     COMMAND
    1  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  systemd --
switched-~tem --deserialize 21
    2  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [kthreadd]
    3  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [ksoftirqd/0]
    5  be/0  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [kworker/0:0H]
    6  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [kworker/u30:0]
    9  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [rcu_sched]
    8  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [rcu_bh]
    9  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [rcuos/0]
   10  be/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [rcuob/0]
   11  rt/4  root        0.00 B/s    0.00 B/s   0.00 %    0.00 %  [migration/0]
```

strace Utility

- The `strace` utility is a debugging tool.
- It prints the system calls made by another program or process.
- Each line contains the system call name, followed by its arguments in parentheses and its return value.
- Errors typically return a value of `-1` and have the `errno` symbol and error string appended.
- Signals are printed as a signal symbol and a signal string.
- Output is printed on standard error or to the file specified with the `-o` option.
- `strace <command>` traces the given `command`.
- `strace -p <PID>` traces the active process given by the process ID (PID).
 - Terminate tracing with CTRL + C.
- `strace -p <PID> -f` traces child processes created by the process being traced with PID.

netstat Utility

- The `netstat` utility displays various network-related information.
- The `netstat` command without options displays a list of open sockets for each address family (AF).
- Several options exist:
 - `-A`: Specify the address family.
 - `-r`: Display the route table.
 - `-i`: Display network interface information.
 - `-s`: Display summary statistics for each protocol.
 - `-g`: Display multicast group membership information.
 - `-n`: Display IP addresses instead of the resolved names.
 - `-c`: Print information every second continuously.
 - `-e`: Display extended information.

netstat Command Alternatives

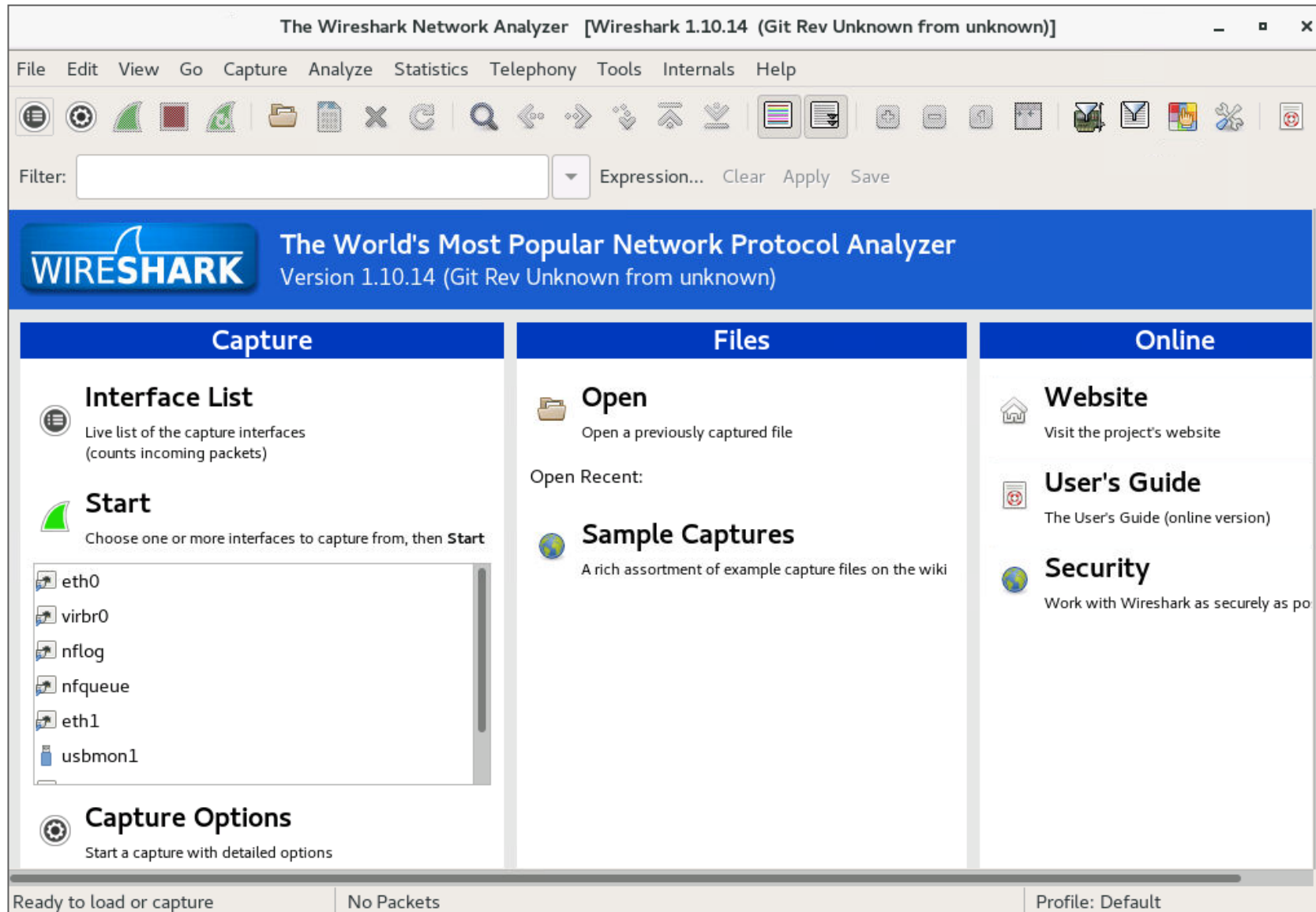
- The `netstat` command is deprecated.
- Some current alternatives for the `netstat` command and options are shown here:

Netstat command	Alternative Command
<code>netstat</code>	<code>ss</code>
<code>netstat -A FAMILY</code>	<code>ss -f FAMILY</code>
<code>netstat -n</code>	<code>ss -n</code>
<code>netstat -r</code>	<code>ip route</code>
<code>netstat -i</code>	<code>ip -s link</code>
<code>netstat -g</code>	<code>ip maddr</code>

tcpdump Utility

- The `tcpdump` utility is a packet-capture utility for network troubleshooting.
- Traffic is captured based on a specified filter.
- A variety of options exist, including:
 - `-D`: Print a list of network interfaces.
 - `-i`: Specify an interface on which to capture.
 - `-c`: Specify the number of packets to receive.
 - `-v`, `-vv`, `-vvv`: Increase the level of detail (verbosity).
 - `-w`: Write captured data to a file.
 - `-r`: Read captured data from a file.
- You can also specify host, source, or destination of traffic and a specific protocol to capture.
- Boolean operators (`AND`, `OR`, `NOT`) allow complex filters.

Wireshark



OSWatcher (OSWbb)

- OSWbb collects and archives operating system and network metrics to aid in diagnosing performance issues.
- OSWbb includes a built-in analyzer called OSWbba.
- Download the OSWbb TAR file from My Oracle Support (MOS).
- To install OSWbb, use the `tar` command:

```
# tar xvf oswbb812.tar
```

- To start OSWbb, use the following command:

```
# ./startOSWbb.sh
```

- To stop OSWbb, use the following command:

```
# ./stopOSWbb.sh
```

OSWatcher (OSWbb)

- List the contents extracted:

```
# ls oswbb
```

analysis	iosub.sh	oswsub.sh
archive	locks	psmemsub.sh
call_du.sh	ltop.sh	sarsub.sh
call_sar.sh	mpsub.sh	src
call_uptime.sh	nfssub.sh	startOSWbb.sh
data	OSWatcherFM.sh	stopOSWbb.sh
docs	OSWatcher.sh	tar_up_full_archive.sh

OSWatcher (OSWbb)

- Start the script and collects data at 60-second intervals and logs the last 10 hours of data.

```
# ./startOSWbb.sh 60 10
# ... Setting the archive log directory to /root/oswbb/archive
Testing for discovery of OS Utilities...
VMSTAT found on your system.
IOSTAT found on your system.
MPSTAT found on your system.
IFCONFIG found on your system.
...
Starting Data Collection...
oswbb heartbeat: date/time
oswbb heartbeat: date/time (60 seconds later)
...
```

OSWbb Diagnostic Data Output

- `OSWatcher.sh` is the main controlling script that spawns other scripts to collect diagnostic data.
- The data is stored in hourly archive files:
 - `<node_name>_<OS_utility>_YY.MM.DD.HH24.dat`
- Data files are located in subdirectories created in the `archive` directory.
- `oswiosstat`: Contains the output from the `iostat` utility
- `oswmeminfo`: The contents of the `/proc/meminfo` file
- `oswmpstat`: Contains the output from the `mpstat` utility
- `oswnetstat`: Contains the output from the `netstat` utility
- `oswprvtnet`: Contains the status of RAC private networks
 - Requires you to manually create an executable file named `private.net`, which runs `traceroute` commands

OSWbb Diagnostic Data Output

- `oswps`: Contains the output from the `ps` utility
- `oswslabinfo`: Contents of the `/proc/slabinfo` file
 - Contains statistics on the kernel slab cache
- `oswtop`: Contains the output from the `top` utility
- `oswvmstat`: Contains the output from the `vmstat` utility
- `oswifconfig`: Contains the output from the `ifconfig` utility
- `oswcputinfo`: Contents of the `/proc/cputinfo` file

OSWatcher Analyzer (OSWbba)

- OSWbba:
 - Is a graphical and analysis utility that is included with OSWbb v4.0.0 and higher
 - Graphically displays data collected and generates reports
 - Includes a built-in analyzer to provide details on performance problems
 - The ability to create a graph and analyze this information relieves you of manually inspecting all the files.
- To start OSWbba, use the following command:

```
# java -jar oswbba.jar -i ~/oswbb/archive
```

- The OSWbba menu provides options to graph and analyze the collected data.

Oracle Enterprise Manager Cloud Control

- System management software that provides centralized monitoring, administration, and life cycle management for IT infrastructures
- Allows you to set up, manage, and support enterprise clouds and traditional Oracle IT environments
- Provides functionality for Oracle Linux server life cycle management
 - Oracle Linux 7.x/8.x/9.x patching is supported from Oracle Enterprise Manager Cloud Control 13c version 13.2.
- Centralized web-based user interface for all management functions
- Select product features:
 - Asset discovery
 - Asset monitoring
 - Oracle Linux provisioning
 - Managing Oracle Linux patching and compliance
 - Running procedures to automate tasks

Oracle Enterprise Manager Cloud Control Components

The following components are part of Oracle Enterprise Manager Cloud Control:

- Oracle Management Agent
- Oracle Management Service (OMS)
- Oracle Management Repository
- Plug-ins
- Enterprise Manager Cloud Control Console
- Oracle JVM D Engine
- Oracle BI Publisher
- EMCTL
- EM CLI

Oracle Enterprise Manager Cloud Control GUI Sections

The screenshot displays the Oracle Enterprise Manager Cloud Control 13c interface. The top navigation bar includes links for Enterprise, Targets, and Favorites. A left-hand menu lists various management areas like Summary, Monitoring, Job, Reports, Configuration, Compliance, Provisioning and Quality Management, My Oracle Support, Cloud, Chargeback, and Consolidation. The main content area shows a 'Welcome to Enterprise Manager Cloud Control 13c' banner, followed by 'Latest Features' and 'Learn More' sections. Below this is a 'Select Enterprise Manager Home Page' section. The bottom section, 'Enterprise Summary', provides an overview of targets' health, including a 'Status' gauge showing 20 targets with a 100% success rate, a 'Compliance Summary' table, and a 'Patch Recommendations' section. Red callout lines point to specific features: 1. Personalized Home Page (top right), 2. Global Menu (top left), 3. Global Menu (top right), 4. Global Search (top center), and 5. Enterprise Summary (bottom center).

1 Personalized Home Page
Enables you to set a particular page as your home page

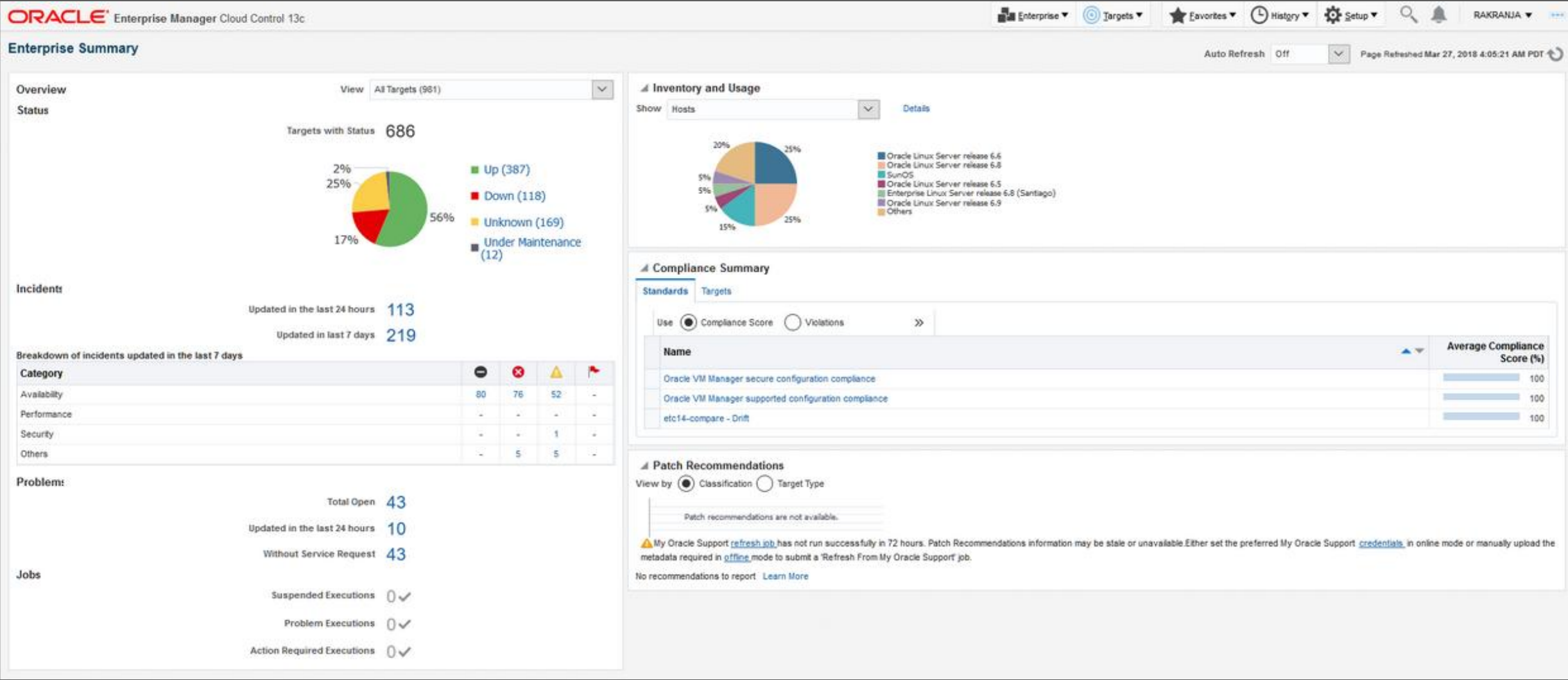
2 Global Menu
Offers access to key functionality

3 Global Menu
Offers access to setup and personalization options

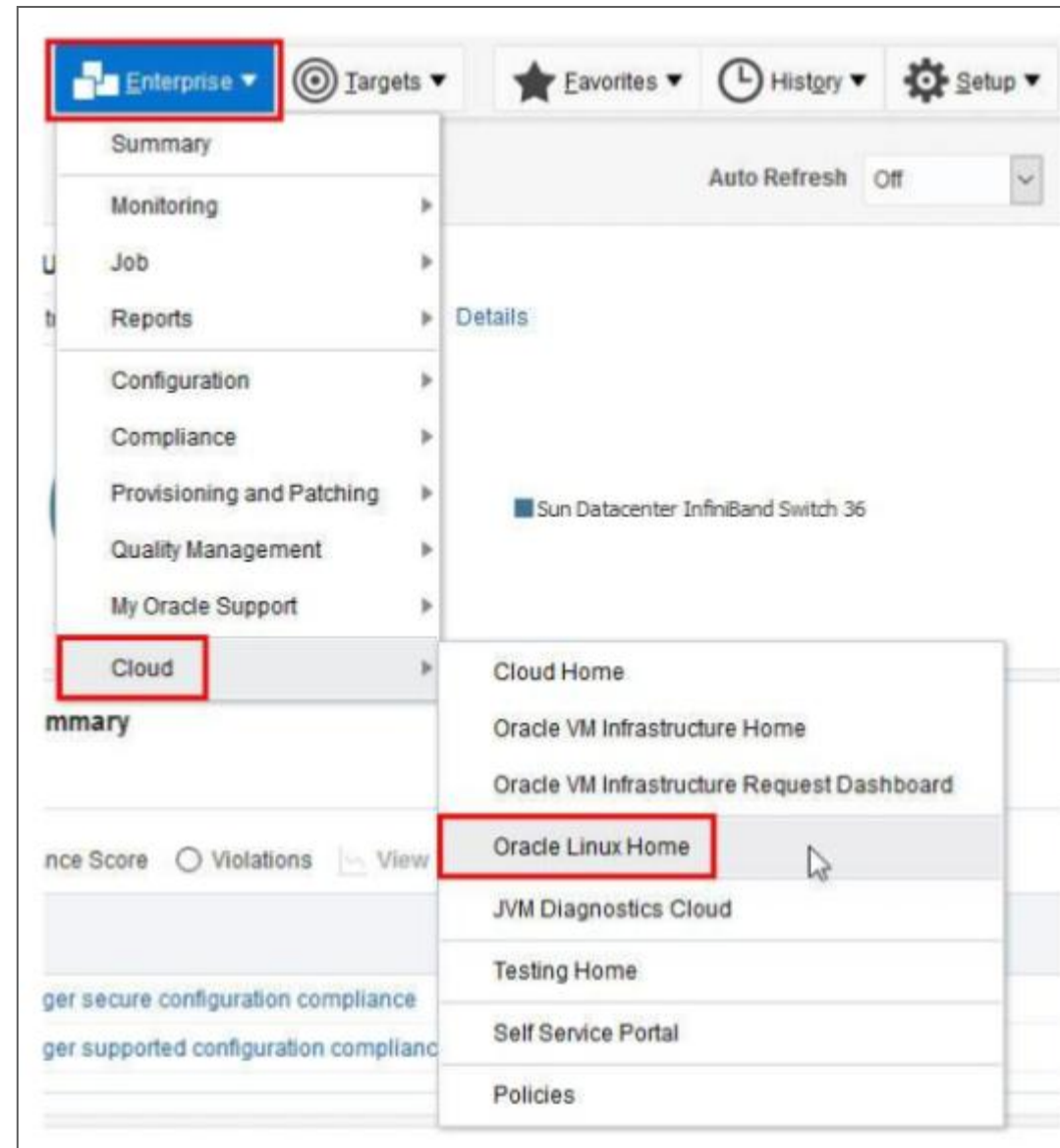
4 Global Search
Enables you to search based on various key search filters

5 Enterprise Summary
Provides an overview of the targets' health

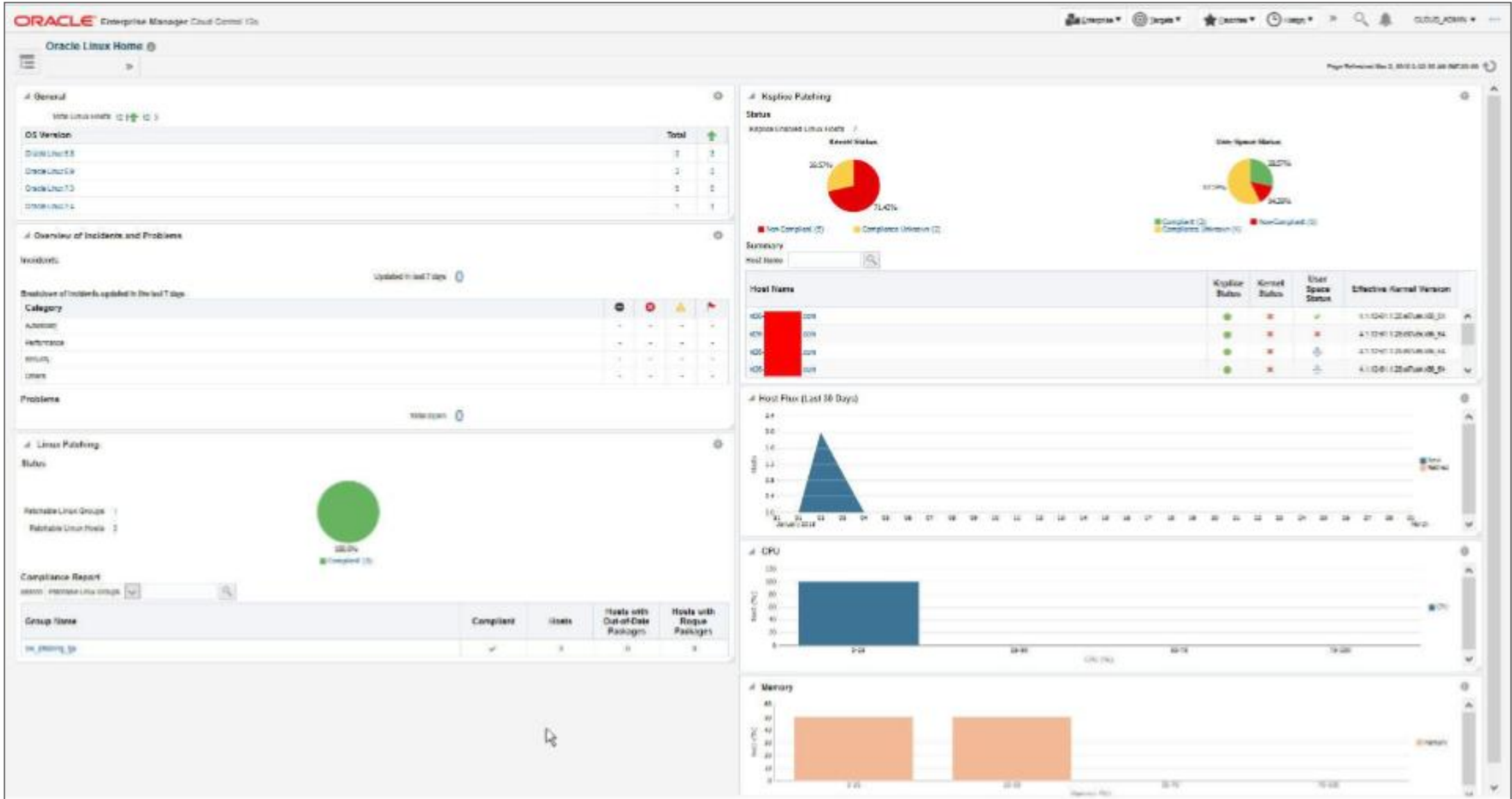
Oracle Enterprise Manager Cloud Control GUI—Enterprise Summary



Oracle Enterprise Manager Cloud Control GUI—Oracle Linux Home



Oracle Linux Home Page



Spacewalk: Overview

- Spacewalk is a full lifecycle management tool for RPM-based Linux distributions.
 - The community project can be found at <https://spacewalkproject.github.io>.
 - Documentation is available at https://docs.oracle.com/cd/E92593_01/index.html.
- Open Source
- Basis for Red Hat Satellite Server 5.x and SUSE Manager
- The RPMs for Oracle Linux are available from the Oracle Linux yum server at yum.oracle.com.
 - Spacewalk Server for Oracle Linux 6 and 9 x86_64
 - Spacewalk Client for Oracle Linux 9 x86_64
 - Spacewalk Client for Oracle Linux 6 i386 and x86_64 architectures
 - Spacewalk Client for Oracle Linux 5 i386 and x86_64 architectures

Spacewalk: Features and Functionality

- Manages software updates
- Allows update staging through multiple environments
- Provides a central web-based administration interface
- Enables scheduling of mass updates to thousands of servers
- Allows delivery of software updates targeting specific errata
- Mirrors content from ULN and Oracle Linux yum locally
- Can manage internal and third-party Yum repositories
- Provides geographic distribution by using Spacewalk Proxy and Inter-Server Synchronization (ISS)

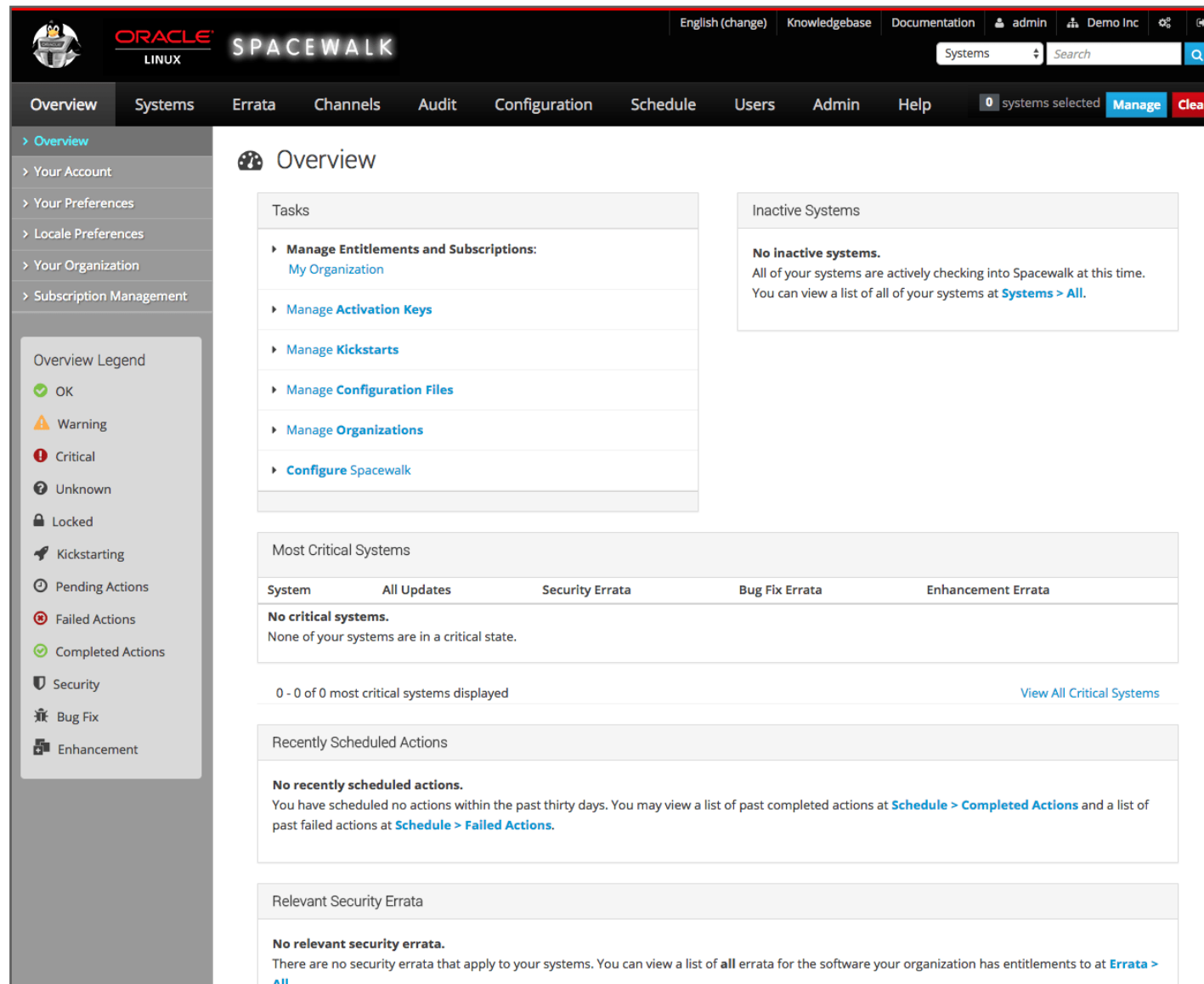


Spacewalk: Features and Functionality

- Provisioning of new physical and virtual servers
 - Supports PXE-based deployments by using kickstart
 - Automatically registers new servers
 - Can redeploy existing servers by using PXE
 - Can create new virtual instances
- Remote Management
 - Spacewalk clients connect with the Spacewalk server every 4 hours checking for any actions to run.
 - OSAD provides near-real time triggering of actions on clients.
 - `rhncfg` provides local configuration file management and remote actions.
- Auditing
 - Spacewalk can trigger OpenSCAP-based XCCDF testing and provide results.



Spacewalk Web Interface - Overview Page



The screenshot displays the Oracle Spacewalk web interface. At the top, the header includes the Oracle Linux logo, the 'SPACEWALK' title, and navigation links for 'English (change)', 'Knowledgebase', 'Documentation', 'admin', 'Demo Inc', and a search bar. Below the header is a main navigation bar with tabs: Overview, Systems, Errata, Channels, Audit, Configuration, Schedule, Users, Admin, and Help. A status bar indicates '0 systems selected' with 'Manage' and 'Clear' buttons.

The left sidebar contains a list of links: Overview, Your Account, Your Preferences, Locale Preferences, Your Organization, and Subscription Management. Below these is an 'Overview Legend' with icons for OK, Warning, Critical, Unknown, Locked, Kickstarting, Pending Actions, Failed Actions, Completed Actions, Security, Bug Fix, and Enhancement.

The main content area is titled 'Overview' and features several sections:

- Tasks:** A list of links including 'Manage Entitlements and Subscriptions: My Organization', 'Manage Activation Keys', 'Manage Kickstarts', 'Manage Configuration Files', 'Manage Organizations', and 'Configure Spacewalk'.
- Inactive Systems:** A message stating 'No inactive systems. All of your systems are actively checking into Spacewalk at this time. You can view a list of all of your systems at [Systems > All](#).'
- Most Critical Systems:** A table with columns: System, All Updates, Security Errata, Bug Fix Errata, and Enhancement Errata. The message below the table states 'No critical systems. None of your systems are in a critical state.' Below the table, it shows '0 - 0 of 0 most critical systems displayed' and a link to 'View All Critical Systems'.
- Recently Scheduled Actions:** A message stating 'No recently scheduled actions. You have scheduled no actions within the past thirty days. You may view a list of past completed actions at [Schedule > Completed Actions](#) and a list of past failed actions at [Schedule > Failed Actions](#).'
- Relevant Security Errata:** A message stating 'No relevant security errata. There are no security errata that apply to your systems. You can view a list of all errata for the software your organization has entitlements to at [Errata > All](#).'

Quiz



Which of the following utilities allows you to collect system information for sending to Oracle support? (Select all that apply.)

- a. `sosreport`
- b. `sar`
- c. `OSWbb`
- d. `strace`



Quiz



Which of the following utilities allows you to collect system information for sending to Oracle support? (Select all that apply.)

- a. `sosreport`
- b. `sar`
- c. `OSWbb`
- d. `strace`



Quiz



Which of the following utilities monitors system processes in real time?

- a. mpstat
- b. iostat
- c. vmstat
- d. top



Summary

In this lesson, you should have learned how to:

- Use the `sosreport` utility
- Use the `iostat`, `mpstat`, `vmstat`, `sar`, `top`, `iostat`, and `strace` utilities
- Use the `netstat` and `tcpdump` utilities
- Use the Wireshark network analyzer GUI
- Use the OSWatcher (OSWbb) tool
- Use OSWatcher Analyzer (OSWbba)
- Describe Oracle Enterprise Manager Cloud Control
- Describe Spacewalk



Practices for Lesson 18: Overview

The practices for this lesson cover the following topics:

- Using `sosreport` to collect system information
- Using standard Linux performance monitoring tools
- Installing and Using OSWatcher
- Using OSWatcher Analyzer